Abhinav Sharma

Applied AI/ML Engineer — LLM Agents · RAG · Backend

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SUMMARY

Applied AI engineer who ships LLM products end to end. Sole engineer on Zavyo, a multi-tenant WhatsApp SaaS with RAG chatbots and tool-calling AI agents, built and deployed in ~5 weeks (the company shut down before public launch). Builds daily with coding agents (Claude Code, OpenCode), including App Foundry, a prompt-to-app platform, and a WhatsApp voice-note CRM running on it. Background in deep learning and computer vision.

EXPERIENCE

Synditech Technologies — SDE 1, Full-Stack & AI Engineer — Dec 2025 – May 2026

Zavyo: multi-tenant WhatsApp marketing & AI conversation SaaS (FastAPI, React 19, MongoDB, Qdrant, OpenAI) · Showcase

- Sole engineer: designed, built and deployed the platform on Meta's WhatsApp Cloud API, from first commit to a deployed, launch-ready build in ~5 weeks (400+ commits)
- Built tool-calling AI agents with per-bot configurable tools (sales handoff, external HTTP calls, product carousel, scheduled follow-ups), plus AI-based lead classification from conversations
- Built a RAG pipeline from scratch: PDF/DOCX/TXT/CSV ingestion, boundary-aware chunking, OpenAI embeddings, a separate Qdrant collection per tenant, and embedding usage metered against tenant credits
- Engineered race-safe, idempotent Razorpay billing: an atomic compare-and-set claim lock prevents double-crediting when webhooks and client verification arrive concurrently, across all payment paths
- Enforced tenant isolation across REST, WebSocket and vector layers with 50+-key RBAC; built a stateful router that runs no-code ReactFlow graphs as live WhatsApp conversations (300+ endpoints, 70+ collections, Dockerized on Coolify)

OPEN SOURCE

LiteLLM (open-source LLM gateway) · PR #45358 — Oct 2026

- Fixed Vercel AI Gateway embedding models being registered as chat models, which broke proxy health checks; fixed the price-sync script and cost maps, added regression tests

PROJECTS

WhatsApp Voice-Note CRM — Meta WhatsApp Cloud API, OpenAI tool calling, speech-to-text, Supabase (Postgres + RLS) · Demo video

- A team updates one shared CRM by sending WhatsApp voice notes or business-card photos; transcription and LLM tool calls turn them into structured records, and reports are pulled from chat; row-level security and an audit trail on every change

App Foundry — prompt-to-app platform (private beta) — coding agents (OpenCode), Docker, Supabase, object storage · Architecture

- A coding agent builds each app inside its own isolated container with a per-app database schema; one-click publish ships built frontends to object storage with instant rollback; idle sandboxes stop and archive automatically and resume on demand

Spoken-English Grammar Scoring (SHL Hiring Assessment, Kaggle) — Whisper, DeBERTa-v3, Qwen2.5-7B, scikit-learn · GitHub

- Scores spoken-English grammar (1–5) from 45–60 s clips: normal + prompted-verbatim Whisper passes (their edit distance is a feature, r = −0.39), CoLA features, an LLM rubric grade and fine-tuned DeBERTa, blended on 5-fold CV (OOF RMSE 0.510; public leaderboard 0.394)

Offline RAG Stack: Air-Gapped AI Deployment — FastAPI, Ollama/vLLM, Qdrant, Docker · GitHub

- Self-hosted RAG system with zero internet egress: all services on an internal Docker network behind a firewalled edge relay (NET_ADMIN-only), with an automated air-gap check proving DNS, TCP and HTTP egress are blocked
- Offline install path (image and model bundle with SHA256 verification), Ollama/vLLM switchable by config, typed 504/502 failure mapping for model and vector-store outages; 77 tests, strict mypy, CI

Quotation Generation Agent — React, FastAPI, OpenAI · GitHub

- LLM agent that turns multi-PDF price sheets and customer emails into structured quotations and flags missing information; dual extraction with automatic fallback, 3-tier JSON recovery, editable agent.md spec, CI

Self-Driving RC Car — Behavioral Cloning — TensorFlow/Keras, OpenCV, Raspberry Pi, FastAPI · GitHub — Dec 2024 – Apr 2025

- Collected ~4,500 labeled frames with a multi-process rig, trained a modified NVIDIA PilotNet CNN (8-class steering), and ran real-time inference over a WebSocket link to the Pi's motor control

SKILLS

AI/ML: LLM agents & tool-calling, RAG, speech-to-text (Whisper), OpenAI API, Ollama, PyTorch, TensorFlow/Keras, Hugging Face Transformers, CNNs, Vision Transformers, OpenCV

Backend: Python, FastAPI, Flask, REST, WebSockets, MongoDB, PostgreSQL, Supabase (RLS), Qdrant, WhatsApp Cloud API · **Frontend:** TypeScript, React, Next.js, Vite

Infra & tools: Docker, Coolify, GitHub Actions, AWS (Lambda, S3), object storage (MinIO), Linux, Git, Claude Code, OpenCode

EDUCATION & LEADERSHIP

Vellore Institute of Technology, Bhopal — B.Tech, Computer Science and Engineering — 2022 – 2026

Leadership: Event Management Lead, Open Source Club (Jan 2024 – Dec 2025) · Lab Setup Lead, Digital Logic Design Lab (Feb 2025)